

Generative AI (AI4009)

Final Exam: Part-I

Date: May 21, 2026

Course Instructor

Dr. Akhtar Jamil

Maximum Time 1

(Hrs):

Total Marks: 40

Total Questions: 1

Roll No

Section

Student Signature

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Part-I MCQ

Maximum time allowed is 1 (one hour). However, you can take Part-II after you complete and return Part-I.

Part-I has 40 MCQs while Part-II has three questions.

[CLO 1-3]

Question No. 1. MCQ [1 x 40 = 40]

Answer the MCQs on the given MCQ Answer Sheet attached at the end of the question paper.

Answers marked on the question paper will not be evaluated.

1. A dense weight matrix in a Transformer layer has dimensions: $d \times k = 768 \times 96$

Full fine-tuning would update all weights. LoRA replaces the update with two low-rank matrices of rank: $r = 12$. The LoRA matrices have sizes: $A \in \mathbb{R}^{r \times k}, B \in \mathbb{R}^{d \times r}$

How many LoRA trainable parameters are required?

- A. 20,736
- B. 73,728
- C. 9,216
- D. 10,368

2. A hospital assistant uses RAG to answer questions from medical reports. The retriever returns documents that contain similar medical terms but are about different patients. The LLM then generates a confident but incorrect answer.

Which is the most likely technical cause of failure?

- A. The LLM was not fine-tuned on medical terminology.
- B. The retriever optimized semantic similarity but failed to preserve document-level identity and metadata constraints.
- C. The vector database used embeddings instead of exact keyword matching.
- D. The model generated a response before receiving the user query.

3. According to the “Born-Again Networks” paradigm in self-distillation, what is the observed relationship between the teacher (T) and successive student generations (Sk) using identical architectures?

- A) Network Accuracy $T > S1 > S2 > Sk$
- B) Network Accuracy $T = S1 = S2 = Sk$
- C) Network Accuracy $T < S1 < S2 < Sk$
- D) Network Accuracy is randomized across generations

4. A UNet receives noisy samples x_t , but the timestep t is not provided to the model. Which failure is most likely?

- A. The model will stop using convolutional layers.
- B. The model will automatically infer the exact timestep from the image size.
- C. The model will become a latent diffusion model by default.
- D. The model cannot know the current noise level and may apply inappropriate denoising strength.

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5. In Stable Diffusion, cross-attention is inserted into the denoising architecture. What is its main technical role?
- A. It allows latent image features to attend to text embeddings during denoising.**
 - B. It replaces the diffusion noise schedule.
 - C. It converts final latents into RGB pixels.
 - D. It removes the need for a UNet.
6. A teacher wants an LLM to prepare class material. Which prompt best reflects the GPT-5.5 style of “define the destination; let the model choose the path”?
- A. “Prepare a lecture-ready explanation of prompt engineering for undergraduate students. Include definition, importance, weak-vs-strong prompt comparison, and common mistakes. Use simple academic English.”**
 - B. “First analyze prompt engineering, then create an outline, then write the lecture, then revise it, then check each line, then rewrite the final answer.”
 - C. “Write something useful about prompt engineering.”
 - D. “Act like a professor and explain prompt engineering.”
7. A denoising diffusion model predicts the noise added to a corrupted sample. For one training example, the actual noise vector is ϵ and predicted noise vector ϵ_θ are given as follows:

$$\epsilon = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \quad \epsilon_\theta = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

Using L2 loss, what is the noise prediction loss?

- A. 3
 - B. 4
 - C. 5
 - D. 6**
8. A Vision Transformer receives a color image of size: $224 \times 224 \times 3$ with patch size: 16×16 . What is the length of each flattened patch before linear projection?
- A. 16
 - B. 256
 - C. 768**
 - D. 196
9. A university wants an assistant that answers questions about admission rules. The rules change every semester, and the assistant must quote the latest official policy document. The university also has no labeled training dataset. Which design is most appropriate?
- A. Full fine-tuning, because the model must memorize the admission rules.
 - B. Prompt engineering only, because prompts can replace external knowledge sources.
 - C. Fine-tuning, because retrieval increases factual accuracy only for image data.
 - D. RAG, because the knowledge is dynamic and must be retrieved from current documents.**
10. A pre-trained LLM is fine-tuned only on a narrow classification dataset. After fine-tuning, it performs well on that classification task but becomes weaker at summarization, question answering, and general reasoning tasks. Which explanation best describes this failure?
- A. The model has learned better generalization across all tasks
 - B. The model has reduced inference latency because of quantization
 - C. The model has experienced catastrophic forgetting due to task-specific weight updates**
 - D. The model has converted hard prompts into soft prompts
11. A teacher model produces logits for three classes in the order: [Class A, Class B, Class C] as follows:
- $$z = [4, 2, 0]$$
- Using temperature: $T = 2$ and values: $e = 2.72$. What is the softened probability of Class A?
- A. 0.362
 - B. 0.665**
 - C. 0.739
 - D. 0.500

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Solution:

After temperature scaling:

$$\frac{z}{T} = [2, 1, 0]$$

Softmax for Class A:

$$\begin{aligned} p_A &= \frac{e^2}{e^2 + e^1 + e^0} \\ p_A &= \frac{7.39}{7.39 + 2.72 + 1} \\ p_A &= \frac{11.11}{11.11} \\ p_A &= 0.665 \\ \boxed{0.665} \end{aligned}$$

12. A student model is not forced to copy the exact feature vector of the teacher for each individual input. Instead, it is trained to preserve relationships among samples. For example, if two inputs are close to each other in the teacher's feature space, the student is encouraged to keep them close in its own feature space. Which type of knowledge is being transferred?
- A. Response-based knowledge
 - B. Feature-based knowledge
 - C. Relation-based knowledge**
 - D. Hard-label knowledge
13. A multimodal RAG system ranks retrieved chunks using: Final Score = 0.6(text score) + 0.4(image score). Two candidate chunks are retrieved:
- $$\begin{aligned} C_1: \text{text} &= 0.70, \text{image} = 0.90 \\ C_2: \text{text} &= 0.85, \text{image} = 0.60 \end{aligned}$$
- Which chunk should be ranked higher?
- A. C_1 , because its weighted final score is higher.**
 - B. C_2 , because its text score is higher.
 - C. C_1 , because its image score is higher.
 - D. Both have exactly the same final score.
14. Which prompt is most likely to suffer from over-prompting?
- A. "Summarize zero-shot prompting in 5 bullet points for beginners."
 - B. "Classify the review as positive, negative, or neutral. Return only the label."
 - C. "Explain few-shot prompting. First define every word, then reason through each sentence, then produce exactly 9 sections, then repeat the definition twice, then avoid all examples, then include examples, then make it short but detailed."**
 - D. "Compare zero-shot and few-shot prompting using a table with three differences."
15. A teacher and student produce probability distributions over four classes:
- $$\begin{aligned} p_T &= [0.50, 0.30, 0.15, 0.05] \\ p_S &= [0.40, 0.35, 0.20, 0.05] \end{aligned}$$
- What is the L2 distillation loss?

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- A. 0.010
- B. 0.0150**
- C. 0.0750
- D. 0.0950

Correct Answer: C

Solution:

$$\begin{aligned}\text{Using: } \mathcal{L}_{KD} &= \sum_{i=1}^4 (p_{Ti} - p_{Si})^2 \\ \mathcal{L}_{KD} &= (0.50 - 0.40)^2 + (0.30 - 0.35)^2 + (0.15 - 0.20)^2 + (0.05 - 0.05)^2 \\ &= (0.10)^2 + (-0.05)^2 + (-0.05)^2 + (0)^2 \\ &= 0.0100 + 0.0025 + 0.0025 + 0 \\ &= 0.0150 \\ &\boxed{0.0150}\end{aligned}$$

16. Two neural networks are trained from scratch. Each network is trained using the ground-truth label and also encouraged to match the probability distribution produced by the other network:

$$\begin{aligned}\mathcal{L}_1 &= \mathcal{L}_{CE}(y, p_1) + KL(p_2 \parallel p_1) \\ \mathcal{L}_2 &= \mathcal{L}_{CE}(y, p_2) + KL(p_1 \parallel p_2)\end{aligned}$$

Which statement best describes this training setup?

- A. It is offline distillation because one network must be fixed before the other begins training.
 - B. It is online distillation because both networks are updated during the same training process.**
 - C. It is cross-modal distillation because both networks always use different input modalities.
 - D. It is model quantization because both networks use reduced numerical precision.
17. Suppose the dimension of keys and queries is increased from $d_k = 16$ to $d_k = 1024$, but the dot products are not scaled before softmax using the new dimension length. Which problem is most likely?
- A. Dot products may become very large, causing softmax saturation and weak gradients.**
 - B. Dot products become extremely small for all tokens.
 - C. Softmax will assign equal probability to all tokens.
 - D. The value matrix cannot be multiplied with attention weights.
18. A model is being used for fact-based question answering in a medical information system. The system should avoid creative wording and should return stable, concise, and predictable answers. Which configuration is most appropriate?
- A. High temperature to increase diversity
 - B. Low temperature to make outputs more deterministic**
 - C. High presence penalty to force topic changes
 - D. Maximum top-p to allow more medical terms
19. A researcher selects very large β_t values for early diffusion steps. Which problem is most likely to occur?
- A. The model will preserve fine-grained image structure for more timesteps.
 - B. The reverse model will no longer require timestep conditioning.
 - C. The image may remain mostly clean for too many steps, making early reverse denoising very easy.

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D. The image will become too noisy too rapidly, making it difficult for the reverse denoising model to predict it correctly.

20. An instructor wants an AI assistant to generate feedback on a student's programming assignment. The assignment includes code, marking criteria, and required feedback style. Which prompt design is most aligned with Claude-style prompting?
- A. "You are a good programming teacher. Check this assignment."
 - B. "Read the code and give feedback in any way you think is useful."
 - C. **"Evaluate the submitted code using clearly separated sections for the task, assignment context, grading criteria, constraints, and required feedback format."**
 - D. "Give a long explanation about the code and mention as many problems as possible."
21. A company wants an AI assistant to generate weekly project status summaries from team updates. The updates contain completed tasks, blockers, pending decisions, and risks. Which prompt design best follows the principle of a strong prompt?
- A. "You are a project expert. Summarize everything properly."
 - B. "Read the updates and write a useful report."
 - C. **"Prepare a weekly project summary for engineering managers. Include completed work, unresolved blockers, pending decisions, and risks. Use short sections and avoid adding information not present in the updates."**
 - D. "Think step by step and write the best possible project report."
22. Technically, why does training an LLM require significantly more GPU VRAM than performing inference with the same model?
- A) Inference requires storing the entire history of the optimizer's momentum states.
 - B) Training requires the use of multiple context windows for every attention head simultaneously.
 - C) Inference models are typically required to be FP64, while training is performed at INT8.
 - D) **Training requires saving intermediate activations for the backward pass and storing gradients for every parameter.**
23. In the Scaled Dot-Product Attention mechanism, why is the scaling factor $1/\sqrt{d_k}$ mathematically applied to the dot product of QK^T prior to the softmax operation?
- A) To normalize the embeddings to a unit variance of 1.0 for cross-entropy stability.
 - B) **To counter the growth of the dot product magnitude in high dimensions, preventing the vanishing gradients.**
 - C) To ensure the resulting attention matrix is symmetric and positive definite for optimized GPU computation.
 - D) To reduce the total memory complexity of the subsequent matrix multiplication with the Value (V) matrix.
24. When transferring knowledge between a large teacher network and a smaller student network, an additional transformation layer is sometimes introduced between their intermediate representations. What is the primary technical reason for this?
- A) **To match incompatible feature dimensions**
 - B) To regularize the student model
 - C) To reduce computational complexity
 - D) To stabilize the optimizer state

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25. Given a BERT-Large architecture with 340 million parameters, calculate the minimum GPU VRAM required to store the model weights using FP32 precision (where 1 parameter = 4 bytes).
- A) **1.36 GB**
 - B) 2.68 GB
 - C) 3.72 GB
 - D) 4.08 GB

26. A user writes a very long prompt containing repeated instructions, conflicting requirements, unnecessary steps, and strict formatting rules that are not relevant to the task. According to the GPT-5.5 prompting style, what is the most likely issue?
- A. **The prompt may become noisy, reducing output quality.**
 - B. The model will always produce a more accurate answer because the prompt is longer.
 - C. The model will automatically ignore all irrelevant instructions.
 - D. The prompt becomes a few-shot prompt because it contains many instructions.

27. Let: $Q \in \mathbb{R}^{6 \times 8}$, $K \in \mathbb{R}^{6 \times 8}$, $V \in \mathbb{R}^{6 \times 10}$. Using scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

What is the final output shape?

- A. 6×6
 - B. 6×8
 - C. **6×10**
 - D. 8×10
28. An AI assistant is asked to process customer complaints using external tools. The system must search only when the complaint contains a product name, cite evidence from retrieved documents, stop when no matching record is found, and report uncertainty instead of guessing. Which concept is most strongly represented?
- A. Role prompting only
 - B. **Agentic and tool-based prompting**
 - C. Zero-shot prompting
 - D. Model quantization
29. Consider a teacher model producing logits [5, 1] for a binary classification task. When the temperature T is increased from 1 to 10 during distillation, which change most directly improves the student's learning signal?
- A) The student receives sharper gradients for the correct class
 - B) The probability mass collapses to the dominant class
 - C) **The relative confidence gap between classes is reduced**
 - D) The cross-entropy loss becomes independent of logits

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30. A Transformer encoder receives the following two token sequences:

Sequence A: "The cat chased the dog"

Sequence B: "The dog chased the cat"

Assume the model is given only token embeddings and no positional encoding. What issue is most likely to occur?

A. The model will still fully distinguish both sequences because self-attention automatically stores word order.

B. The model may confuse the two sequences because self-attention alone does not explicitly encode token position.

C. The model will stop computing query, key, and value matrices.

D. The model will behave exactly like an LSTM.

31. A query vector and three key vectors are given and $d_k = 4$:

$$Q = [2, 0, 1, 1] \quad K_1 = [1, 0, 1, 0], \quad K_2 = [0, 1, 1, 1], K_3 = [2, 0, 0, 1]$$

Which key receives the highest scaled attention score?

A. K_1

B. K_2

C. K_3

D. All keys receive equal score

Solution:

$$QK_1^T = (2)(1) + (0)(0) + (1)(1) + (1)(0) = 3$$

$$QK_2^T = (2)(0) + (0)(1) + (1)(1) + (1)(1) = 2$$

$$QK_3^T = (2)(2) + (0)(0) + (1)(0) + (1)(1) = 5$$

Since:

$$\sqrt{d_k} = \sqrt{4} = 2$$

scaled scores are:

$$K_1 = 1.5, K_2 = 1, K_3 = 2.5$$

So K_3 receives the highest score.

32. A deep Transformer is trained without residual connections. Which problem is more likely to occur?

A. The model becomes easier to optimize because gradients no longer pass through shortcuts.

B. Gradient flow may become weaker, making deeper layers harder to train stably.

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- C. Positional encoding becomes unnecessary.
- D. Softmax becomes linear.

33. Assume two adaptation strategies are considered for a 10B-parameter language model across 20 downstream tasks:

- Strategy X: Full fine-tuning per task
- Strategy Y: LoRA-based adaptation per task

Which one of the following statements is most technically accurate and suitable for this scenario?

- A) Strategy X is preferred as it minimizes catastrophic forgetting across tasks
- B) Strategy X is preferable when inference latency is not critical
- C) Strategy Y is preferred as it reduces parameter storage while preserving previous knowledge**
- D) Strategy Y eliminates the need for task-specific supervision

34. A model is instruction-tuned using (instruction, input, output) triplets rather than task-specific labels. What is the primary benefit of this approach?

- A) Reduced training time due to fewer parameters
- B) Improved instruction following and better generalization across diverse tasks**
- C) Elimination of the need for evaluation metrics
- D) Guaranteed robustness to distribution shift

35. In Knowledge Distillation, the Temperature (T) parameter is applied to the modified softmax function. What is the specific effect of setting $T > 1$?

- A) It generates a smoother probability distribution.**
- B) It produces a sharper distribution.
- C) It shifts all logit values into a negative range to facilitate faster L2 loss convergence.
- D) It forces the student model to ignore the logits and learn exclusively from the teacher's hint layers.

36. In Prompt Engineering, how does Top P (Nucleus) sampling technically restrict the token selection pool?

- A) It masks out any token with a probability lower than a fixed constant P.
- B) It enforces a frequency penalty on tokens that have already appeared in the current sequence.
- C) It divides the logits by a factor of P to increase the diversity of the output.
- D) It limits the sampling pool to the smallest set of tokens whose cumulative probability exceeds the threshold P.**

37. In Multi-Teacher Knowledge Distillation, how is the collective knowledge typically synthesized to guide the student model?

- A) The student trains on a different subset of the dataset for each individual teacher model.
- B) Only the teacher with the highest KL-divergence relative to the student is used for the loss calculation.
- C) The student mimics the average response (logits or feature representations) across the ensemble of teachers.**
- D) The teacher models are trained online while the student model's parameters remain fixed.

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38. In the BERT architecture, what is the specific technical role of the [CLS] token?
- A) To act as a delimiter separating Sentence A from Sentence B in the input stream.
 - B) To encode the positional index of the first token in the 3D embedding space.
 - C) To provide a target for the Masked Language Modeling task in the hidden layers.
 - D) To aggregate information from all tokens for sequence-level classification.**
39. A Transformer-based language model has a vocabulary size of 50,000 tokens. During autoregressive text generation, the model predicts the next token at each time step. How many probability values are produced by the model's final softmax layer for each next-token prediction?
- A) 512
 - B) 1
 - C) 50,000**
 - D) Equal to the number of attention heads
40. DeepSeek-V3 utilizes an auxiliary-loss-free load balancing strategy. If a specific expert i is determined to be overloaded after a training batch, how is the routing mechanism adjusted?
- A) The expert is temporarily removed from the Top-K selection for the duration of the next epoch.
 - B) The softmax temperature for that expert's gating score is increased to flatten the selection probability.
 - C) The router's weight matrix for that specific expert is increased by a hyperparameter γ .
 - D) The routing bias term b_i for that expert is decreased by a factor of γ .**

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41. A large language model is fine-tuned on a narrow domain task and shows strong performance on that task but degraded performance on previously learned tasks. Which mechanism best explains this behavior?
- A) Overfitting caused by insufficient regularization
 - B) Excessive gradient updates overriding previously learned representations**
 - C) Vanishing gradients in lower layers
 - D) Inference-time distribution shift
42. Why is feature-based knowledge distillation often preferred over response-based distillation when the student architecture differs significantly from the teacher?
- A) Feature-based distillation does not require labels
 - B) Logits cannot be computed for heterogeneous architectures
 - C) Response-based distillation requires identical output layers
 - D) Intermediate representations encode transferable structural information**
43. Two neural networks are trained simultaneously, each using ground-truth labels while also learning from the predictions of each other. Which loss formulation is most suitable to enable effective mutual knowledge transfer during training?
- A) Mean squared error applied only to the final logits
 - B) Cross-entropy loss with hard labels only
 - C) A combination of cross-entropy loss and a KL-divergence loss**
 - D) Hinge loss applied to intermediate feature representations
44. During BERT pre-training, the Masked LM (MLM) task involves a specific replacement strategy for the 15% of tokens selected for prediction. What is generally more appropriate technical distribution of this strategy?
- A) 100% replacement with the [MASK] token to maximize the bidirectional learning objective.
 - B) 80% replacement with [MASK] and 20% replacement with the [SEP] token to facilitate Next Sentence Prediction.
 - C) 50% replacement with [MASK], 25% replacement with a random token, and 25% remains unchanged.
 - D) 80% replacement with [MASK], 10% replacement with a random token, and 10% remains unchanged.**
45. In Variational Autoencoders (VAEs), the Reparameterization Trick is implemented to resolve which fundamental structural training limitation?
- A) The vanishing gradient problem inherent in extremely deep residual encoder layers.
 - B) The inability to backpropagate gradients through a stochastic sampling.**
 - C) The high computational complexity of calculating the KL-Divergence analytically for non-Gaussian priors.
 - D) The tendency of the latent space to collapse into a single point during Evidence Lower Bound (ELBO) optimization.
46. Within the Model Context Protocol (MCP), which feature allows an LLM to actively access data from databases or call external APIs?
- A) Resources
 - B) Prompts
 - C) Tools**
 - D) Roots
47. Chain-of-Thought (CoT) prompting is equally effective for small and large language models, since reasoning quality depends only on the prompt structure and not on model capacity.
- A) True
 - B) False**

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48. How does LoRA (Low-Rank Adaptation) maintain parameter efficiency while adapting a frozen Large Language Model?
- A) **By injecting and training two small rank decomposition matrices while keeping the original pre-trained weights frozen.**
 - B) By selective pruning of non-essential weights in the Feed-Forward layers of the transformer.
 - C) By adding soft prompts to the input embedding layer that act as the only trainable parameters.
 - D) By quantizing the entire model to 4-bit precision and training only the quantization scale factors.
49. The DeepSeek-V3 architecture utilizes a Mixture-of-Experts (MoE) configuration. For a model with 671B total parameters, what is the activation count per token?
- A) All 671B parameters are activated for every token to maximize reasoning depth.
 - B) **It activates 37B parameters for each token to maintain computational efficiency.**
 - C) It activates only the shared expert Ns unless the router detects a complex query.
 - D) Parameters are activated based on a 4-of-64 expert configuration, totaling 236B parameters.
50. Chain-of-Thought (CoT) prompting is categorized as an emergent ability because:
- A) It is hard coded into the attention masks of modern Transformer architectures.
 - B) **It significantly improves performance primarily as model scale increases**
 - C) It allows the model to update its internal weights during the inference pass via backpropagation.
 - D) It relies on the model's ability to use Scratchpads for storage outside the context window.
51. Suppose you are designing a large-scale Multimodal RAG system for an enterprise knowledge base containing millions of images and documents. To reduce inference latency, the system applies aggressive embedding compression and combines modality-specific retrieval results only at the final response stage, without enforcing tight cross-modal coupling during representation learning.
- Which risk is most likely to increase as a direct consequence of these design decisions?
- A) **Semantic mismatch between retrieved visual and textual information**
 - B) Text extraction failures caused by degraded optical character recognition
 - C) Increased sensitivity to background artifacts present in images
 - D) Higher computational overhead during offline embedding generation
52. When implementing a RAG retriever, Semantic Embeddings are technically preferred over Sparse Embeddings because:
- A) Sparse embeddings are computationally more expensive to calculate due to high-dimensional zero-vectors.
 - B) **Semantic embeddings capture deeper contextual meanings rather than relying on exact lexical matching.**
 - C) Sparse embeddings are limited to a context window of 128 tokens.
 - D) Semantic embeddings use the [CLS] token to bypass the vector database search entirely.
53. The Self-Consistency technique improves CoT reasoning by:
- A) Forcing the model to generate the exact same answer path every time using greedy decoding.
 - B) Penalizing the model if its output exceeds the context window's token limit.
 - C) Training a secondary verifier model to check the logic of each intermediate step in real-time.
 - D) **Sampling a diverse set of reasoning paths and selecting the final answer via a majority vote of consistent outputs.**
54. A RAG-based question-answering system consistently returns fluent but factually incorrect answers, even though relevant documents exist in the vector database. Which component is MOST likely responsible for this failure?
- A) **Ineffective chunking leading to loss of fine-grained semantic matches**
 - B) Use of cosine similarity instead of Euclidean distance

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- C) Over-parameterization of the generator LLM
- D) Excessive prompt length exceeding the LLM context window

55. Quantizing a model from FP32 to INT8 precision generally results in which technical outcome?
- A) A 25% reduction in memory usage with a linear increase in latency.
 - B) A 75% reduction in memory usage and faster integer operations on modern hardware.**
 - C) A 50% increase in model accuracy due to the reduction of floating-point noise.
 - D) The model becomes incompatible with standard Transformer attention mechanisms.
56. Quantizing a BERT-Base model (110M parameters) from FP32 (4 bytes per parameter) to INT8 (1 byte per parameter) results in a memory saving of how many megabytes?
- A) 110 MB
 - B) 220 MB
 - C) 330 MB**
 - D) 440 MB
57. A system must answer questions about rapidly changing financial data (e.g., daily stock prices) while maintaining transparency by referencing original documents. Which approach is MOST appropriate?
- A) Fine-tuning the LLM weekly with updated data
 - B) Using RAG with a non-parametric external knowledge store**
 - C) Increasing model size to reduce hallucinations
 - D) Applying self-distillation on financial reports
58. In a Nucleus (Top-p) sampling strategy with $p = 0.80$, five tokens have the following probabilities: A (0.45), B (0.30), C (0.15), D (0.05), and E (0.05). Tokens are sorted in descending order of probability and added until the cumulative probability reaches or exceeds p . Which tokens are included in the nucleus set?
- A) Tokens A only
 - B) Tokens A and B
 - C) Tokens A, B, and C**
 - D) Tokens A, B, C, D, and E
59. An organization is evaluating the use of an Agentic AI system that performs multi-step reasoning, tool invocation, and environment interaction. In which of the following scenarios would an agent-based approach be least appropriate, despite the availability of high-quality models?
- A) A compliance monitoring system that periodically queries regulations, checks policy updates, and revises recommendations over time
 - B) A conversational data-analysis assistant that iteratively queries databases and refines hypotheses based on intermediate results
 - C) A latency-critical signal processing pipeline that must produce deterministic outputs within microsecond-level deadlines**
 - D) A decision-support agent that explores alternative strategies, executes simulations, and adapts actions based on observed outcomes
60. A production AI assistant dynamically selects relevant conversation history, retrieves user-specific preferences, queries external documents, and invokes tools before forming the final prompt sent to the language model. The system is designed to support many tasks rather than a single fixed prompt. Which concept best describes this approach?
- A) Prompt engineering
 - B) In-context learning
 - C) Fine-tuning
 - D) Context engineering**
61. Based on relation-based knowledge distillation using Maximum Mean Discrepancy (MMD), if the average teacher feature value is 2.33 and the average student feature value is 2.20, what is the MMD value?

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- A) 0.0030
- B) 0.2800
- C) 0.0169**
- D) 0.1178

62. In a Transformer architecture utilizing Multi-Head Attention with $d_{\text{model}} = 512$ and $h = 8$ heads, what are the specific dimensions of the weight matrices W^Q , W^K , and W^V for each individual head?
- A) 512×512
 - B) 512×64**
 - C) 64×512
 - D) 64×64
63. A company must frequently update factual information (e.g., policies, manuals) without retraining its language model. The system must also allow explicit inspection of the source documents used to answer queries. Which design choice and knowledge type best satisfies these requirements?
- A) Fine-tuning the model, relying on parametric knowledge only
 - B) Fine-tuning combined with weight freezing, relying on semi-parametric knowledge
 - C) Retrieval-Augmented Generation using non-parametric knowledge accessible at inference time**
 - D) Pretraining from scratch with larger model capacity to improve memorization
64. DeepSeek-V3 utilizes a Mixture-of-Experts (MoE) gating mechanism where the scores for selected experts are normalized into probabilities (gates). If the top-2 routed experts for a token have logit scores of $s_1 = 2.0$ and $s_2 = 1.0$, what is the approximate normalized gating score g_1 for the first expert? (Assume $e^2 \approx 7.39$ and $e^1 \approx 2.72$)
- A) 0.500
 - B) 0.667
 - C) 0.739**
 - D) 0.269
65. In the Model Context Protocol (MCP), a single MCP client can maintain one shared connection and simultaneously interact with multiple MCP servers without requiring separate client instances.
- A) False**
 - B) True
66. When implementing Multimodal RAG, how does the Early Fusion approach differ technically from Late Fusion?
- A) Late fusion merges embeddings before passing them to the LLM
 - B) Early fusion merges embeddings of different modalities before they are processed by the fusion model**
 - C) Early fusion only supports text-to-text retrieval
 - D) Late fusion requires a shared embedding space like CLIP
67. In a Transformer model with $d_{\text{model}} = 1024$ and 16 attention heads, what is the value of the scaling factor used in the Scaled Dot-Product Attention mechanism?
- A) 4
 - B) 8**
 - C) 16
 - D) 32
68. In the paper titled “Attention Is All You Need”, how was English Constituency Parsing used?
- A) As a pre-training objective for learning syntactic structures.
 - B) As an unsupervised evaluation of word embeddings.
 - C) As a fine-tuning task to improve its translation capabilities.
 - D) As a fine-tuning task to test the Transformer’s ability to generalize beyond translation.**
69. During BERT’s fine-tuning stage, what happens to the model parameters?

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- A) Only task-specific layers are trained
- B) Parameters are re-initialized before training
- C) Encoder layers remain frozen
- D) All pre-trained parameters are updated using labeled downstream data**

70. In the ReAct prompting framework, an agent follows a structured reasoning cycle when solving a task. Which sequence best represents this cycle?

- A) Action → Thought → Observation → Answer
- B) Observation → Thought → Action → Planning → Output
- C) Thought → Observation → Action → Memory Update
- D) Thought → Action → Observation → Reflection → Repeat**

71. In a Vision Transformer (ViT), image patches are flattened and passed through a linear projection before being processed by the Transformer encoder. What is the primary purpose of this linear embedding step?

- A) To convert image patches into fixed-size, lower-dimensional representations suitable for self-attention**
- B) To introduce spatial convolutional inductive bias into the model
- C) To perform nonlinear feature extraction from raw pixel values
- D) To reduce the number of attention heads required in the Transformer

72. An organization uses a large language model to perform a newly defined text classification task. The model is not retrained. Instead, the prompt includes a natural-language task description and one carefully chosen input–output demonstration, after which the model is asked to classify unseen inputs.

Which in-context learning strategy is being applied in this scenario?

- A) Zero-shot learning
- B) One-shot learning**
- C) Few-shot learning
- D) Fine-tuning-based learning

73. In a language model, decoder outputs are decomposed into a reasoning path r_i and a final answer a_i . Which of the following cases demonstrates a minor commonsense reasoning inconsistency, where the reasoning appears plausible but does not logically support the final answer?

A)

Q: Would a glass cup crack if dropped onto concrete?

r_i : Concrete is a rigid surface that can transfer high impact forces to fragile objects.

a_i : Yes, the glass cup would likely crack.

B)

Q: Would a candle continue burning without oxygen?

r_i : Combustion requires oxygen to sustain a flame.

a_i : No, the candle would not continue burning.

C)

Q: Would a metal spoon feel cold at room temperature?

r_i : Metal conducts heat efficiently, allowing it to transfer heat away from the skin faster than other materials.

a_i : No, the metal spoon would feel warm to the touch.

D)

Q: Would wet clothes dry faster on a windy day?

r_i : Wind increases evaporation by removing moist air from the surface of the clothes.

a_i : Yes, wet clothes dry faster on a windy day.

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Rough Work

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Answer Sheet MCQs

Fill the correct option. Only one option must be selected. Selection of multiple options or overwriting will result in ZERO marks.

CORRECT METHOD				WRONG METHOD			
☐ ☒ ☐ ☐				☒ ☐ ☒ ☒			
NAME:				ROLL NO:			
Roll No A B C D							
0 ☐ ☐ ☐ ☐ 1 ☐ ☐ ☐ ☐ 2 ☐ ☐ ☐ ☐ 11 ☐ ☐ ☐ ☐ 12 ☐ ☐ ☐ ☐ 13 ☐ ☐ ☐ ☐							
3 ☐ ☐ ☐ ☐ 4 ☐ ☐ ☐ ☐ 5 ☐ ☐ ☐ ☐ 6 ☐ ☐ ☐ ☐ 7 ☐ ☐ ☐ ☐ 14 ☐ ☐ ☐ ☐ 15 ☐ ☐ ☐ ☐ A B C D							
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10 ☐ ☐ ☐ ☐ 24 ☐ ☐ ☐ ☐ 25 ☐ ☐ ☐ ☐							

Generative AI (AI4009)

Final Exam Part-II

Date: May 21, 2026

Course Instructor

Dr. Akhtar Jamil

Maximum Time (Hrs): 2

Total Marks: 60

Total Questions: 4

Roll No

Section

Student Signature

Do not write below this line

Part-II

Attempt all questions.

[CLO 1-3]

Question No 2. [3 x 10=30]

Write short answers to the following questions.

- I. What may happen if a student model is trained only with hard labels and the teacher model's soft probability distribution is not used during knowledge distillation?

Answer:

The student may learn only the correct class label and may not capture the teacher's uncertainty or class relationship information.

The teacher's soft probabilities show how strongly the teacher relates the input to different classes. Without these probabilities, the student receives a weaker training signal and may learn less effective decision boundaries.

- II. A research team is considering three student-training strategies:

Case No. 1 A large model is first trained completely. Its parameters are then kept fixed, and a smaller model learns from its outputs.

Case No. 2 Two models are trained at the same time, and each model learns partly from the other model's output distribution.

Case No. 3. A single model uses knowledge from its deeper layers or earlier training stages to improve its own shallower layers or later stages.

Please write the name of the distillation strategy that must be employed in each of the above case.

1. Offline
2. Online
3. Self distillation

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- III. Consider a generated sentence has the following precisions: $P_1 = 0.80$, $P_2 = 0.50$ and the candidate length is: $c = 8$. The reference sentence length is: $r = 10$. What is the final BLEU-2 score? (use $e=2.72$)

Solution:

$$\text{Geometric Mean} = \sqrt{P_1 P_2}$$
$$BP = e^{(1-r/c)}$$

Geometric mean:

$$\sqrt{0.80 \times 0.50} = \sqrt{0.40} = 0.632$$

Brevity penalty:

$$BP = e^{(1-10/8)}$$
$$BP = e^{(1-1.25)}$$
$$BP = e^{-0.25} = 0.779$$

BLEU-2:

$$BLEU = BP \times \text{Geometric Mean}$$
$$BLEU = 0.779 \times 0.632$$
$$BLEU = 0.492$$

0.492

- IV. A company wants to deploy a large AI model on a mobile device with limited memory and low computational power. The model's architecture is already suitable for the task, and the company does not want to train a new smaller model. However, it needs to reduce memory usage and improve inference speed.

Which model optimization technique should the company choose: knowledge distillation or quantization? Justify your choice. Also explain a scenario in which the other technique would be more suitable.

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Answer:

The company should choose quantization because it wants to keep the same model architecture but reduce memory usage and improve inference speed. Quantization reduces the precision of model parameters, such as converting FP32 weights to INT8, which makes the model smaller and faster for deployment on limited hardware.

Knowledge distillation would be more suitable if the company wanted to train a smaller student model to imitate a larger teacher model. This would be useful when the original model is too large for deployment and the company is willing to train a compact model while preserving much of the teacher model's performance.

V. Differentiate hard prompting and soft prompting in terms of their working.

Answer:

Hard prompting uses human-readable text instructions written directly into the prompt. These instructions are fixed unless manually changed.

Soft prompting uses trainable embedding vectors that are prepended to the input. These vectors are learned through training and are not necessarily human-readable. The base LLM remains frozen, while only the soft prompt embeddings are updated.

VI. During fine-tuning, a model performs well on a new task but loses performance on previously learned tasks. How can Elastic Weight Consolidation help reduce this problem?

Answer: Elastic Weight Consolidation reduces catastrophic forgetting by restricting large changes to parameters that are important for previously learned tasks. Instead of allowing all weights to change freely during fine-tuning, EWC penalizes changes to important parameters, helping the model adapt to the new task while preserving older knowledge.

VII. Why is the noise prediction objective commonly used in denoising diffusion models instead of training the network to directly reconstruct the clean image?

Answer: The noise prediction objective is useful because the noise added during the forward diffusion process is known. Therefore, the model has a clear supervised target at each timestep.

The model predicts: $\epsilon_{\theta}(x_t, t)$ and compares it with the actual noise: ϵ

using a loss such as: $L(\theta) = \| \epsilon - \epsilon_{\theta}(x_t, t) \|^2$

This helps the model learn how to remove noise step by step during the reverse denoising process. It also makes the training objective more direct and stable because the model learns the corruption component rather than trying to recover the clean image in one step.

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VIII. What may happen if shared experts are removed from a Mixture-of-Experts model?

Answer: If shared experts are removed, the model may lose a common processing path for general knowledge. Routed experts may then repeatedly learn the same basic patterns, such as grammar, common language structure, and simple reasoning.

This can reduce the efficiency of specialization because routed experts may spend capacity learning common information instead of focusing on specific tasks or patterns. As a result, expert capacity may be wasted and the overall MoE model may become less efficient.

IX. Explain the concept of offsets in Parameter-Efficient Fine-Tuning of Transformer-based models.

Answer: Offsets are small trainable updates added to selected weights of a pre-trained Transformer model. Instead of updating the full model, the original weights remain frozen and only these small update components are trained. These offsets represent a compact, low-rank approximation of larger weight changes, allowing the model to adapt to a new task with far fewer trainable parameters.

X. What may be the effect of removing the KL-divergence term from mutual learning?

Answer: Removing the KL-divergence term can weaken the interaction between the models. Each model may learn mainly from the ground-truth labels and may not benefit from the soft prediction patterns of the other model.

Long Questions

Question No 3. [5+5]

- a) Explain the working of CLIP for zero-shot visual classification using natural language supervision. Also draw a neat labeled diagram showing the main components and flow of information in the CLIP zero-shot classification process.

Answer

CLIP, or Contrastive Language-Image Pre-training, is a multimodal model that learns the relationship between images and natural language. It is trained on large-scale image-text pairs, where each image is associated with a natural language description. Instead of learning only fixed class labels, CLIP learns how visual information and text descriptions correspond to each other.

CLIP uses two main components: an image encoder and a text encoder. The image encoder takes an input image and converts it into a numerical representation called an image embedding. The text encoder takes natural language text, such as a caption or a class prompt, and converts it into a text embedding. Both embeddings are mapped into a shared embedding space so that image and text representations can be compared directly.

During training, CLIP uses contrastive learning. For a batch of image-text pairs, the correct image-text pairs are treated as positive pairs, while mismatched image-text pairs are treated as negative pairs. The model is trained to increase the similarity between matching image and text embeddings and

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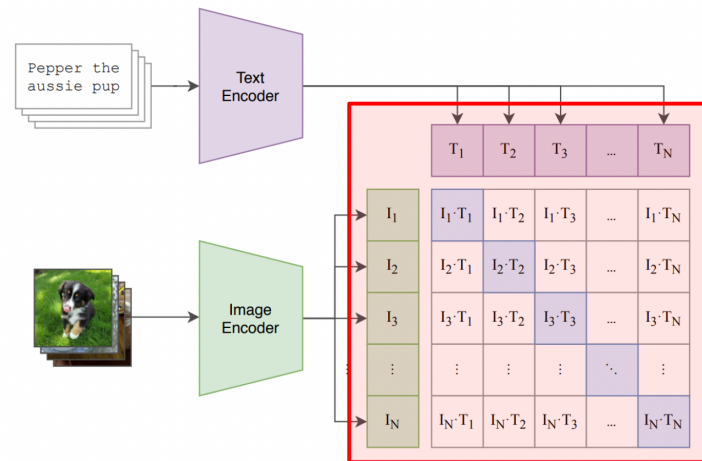
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decrease the similarity between non-matching embeddings. In this way, CLIP learns which textual descriptions are semantically related to which images.

For zero-shot classification, CLIP does not require a newly trained classifier head. Instead, the possible class labels are converted into natural language prompts. For example, if the target classes are “cat,” “dog,” and “car,” they may be written as prompts such as “a photo of a cat,” “a photo of a dog,” and “a photo of a car.” These prompts are passed through the text encoder to generate text embeddings.

The input image is passed through the image encoder to generate an image embedding. CLIP then compares the image embedding with each text embedding using a similarity measure. The prompt with the highest similarity score is selected, and its corresponding class becomes the predicted label. This allows CLIP to classify unseen categories without task-specific training.

CLIP generalizes well because it learns from natural language supervision rather than only fixed class IDs. Natural language prompts provide flexible descriptions of visual concepts. As a result, CLIP can transfer its learned image-text understanding to many visual tasks, such as object recognition, OCR-related understanding, action recognition, and other image classification tasks.



- b) A CLIP model encodes one image and three text prompts into the same embedding space. The image embedding is: $I = [2, 1, 3]$

The text embeddings are:

$$\begin{aligned} T_1 &= \text{"a photo of a cat"} = [1, 1, 2] \\ T_2 &= \text{"a photo of a car"} = [2, 0, 3] \\ T_3 &= \text{"a photo of a tree"} = [0, 2, 1] \end{aligned}$$

Using the absolute difference distance:

$$D(I, T) = \sum_i |I_i - T_i|$$

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find the text prompt that best matches the image.

Answer

Compute the distance between the image embedding and each text embedding.

For T_1 :

$$\begin{aligned} D(I, T_1) &= |2 - 1| + |1 - 1| + |3 - 2| \\ D(I, T_1) &= 1 + 0 + 1 = 2 \end{aligned}$$

For T_2 :

$$\begin{aligned} D(I, T_2) &= |2 - 2| + |1 - 0| + |3 - 3| \\ D(I, T_2) &= 0 + 1 + 0 = 1 \end{aligned}$$

For T_3 :

$$\begin{aligned} D(I, T_3) &= |2 - 0| + |1 - 2| + |3 - 1| \\ D(I, T_3) &= 2 + 1 + 2 = 5 \end{aligned}$$

The smallest distance is:

$$D(I, T_2) = 1$$

Therefore, the image is most similar to:

“a photo of a car”

Question No 4. [5+5]

- a) Consider the following clean 2×2 grayscale image, where pixel values are normalized between 0 and 1:

$$x_0 = \begin{bmatrix} 0.2 & 0.5 \\ 0.7 & 1.0 \end{bmatrix}$$

Assume the diffusion noise schedule is: $\beta_1 = 0.2, \beta_2 = 0.2, \beta_3 = 0.2$

and the Gaussian noise sample is: $\epsilon = \begin{bmatrix} 0.2 & -0.2 \\ 0.1 & -0.1 \end{bmatrix}$

Compute the noisy image at timestep: $t = 3$

Answer

$$\begin{aligned} \alpha_t &= 1 - \beta_t \\ \bar{\alpha}_t &= \prod_{i=1}^t \alpha_i \\ x_t &= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon \end{aligned}$$

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First compute:

$$\begin{aligned}\alpha_1 &= 1 - \beta_1 = 1 - 0.2 = 0.8 \\ \alpha_2 &= 1 - \beta_2 = 1 - 0.2 = 0.8 \\ \alpha_3 &= 1 - \beta_3 = 1 - 0.2 = 0.8\end{aligned}$$

Now compute:

$$\begin{aligned}\bar{\alpha}_3 &= \alpha_1 \alpha_2 \alpha_3 \\ \bar{\alpha}_3 &= 0.8 \times 0.8 \times 0.8 \\ \bar{\alpha}_3 &= 0.512\end{aligned}$$

Next:

$$\begin{aligned}\sqrt{\bar{\alpha}_3} &= \sqrt{0.512} \approx 0.716 \\ \sqrt{1 - \bar{\alpha}_3} &= \sqrt{1 - 0.512} = \sqrt{0.488} \approx 0.699\end{aligned}$$

Now use: $x_3 = \sqrt{\bar{\alpha}_3} x_0 + \sqrt{1 - \bar{\alpha}_3} \epsilon$

Substitute the values: $x_3 = 0.716 \begin{bmatrix} 0.2 & 0.5 \\ 0.7 & 1.0 \end{bmatrix} + 0.699 \begin{bmatrix} 0.2 & -0.2 \\ 0.1 & -0.1 \end{bmatrix}$

First compute:

$$0.716x_0 = \begin{bmatrix} 0.716(0.2) & 0.716(0.5) \\ 0.716(0.7) & 0.716(1.0) \end{bmatrix} = \begin{bmatrix} 0.1432 & 0.3580 \\ 0.5012 & 0.7160 \end{bmatrix}$$

Now compute:

$$0.699\epsilon = \begin{bmatrix} 0.699(0.2) & 0.699(-0.2) \\ 0.699(0.1) & 0.699(-0.1) \end{bmatrix} = \begin{bmatrix} 0.1398 & -0.1398 \\ 0.0699 & -0.0699 \end{bmatrix}$$

Now add both matrices:

$$\begin{aligned}x_3 &= \begin{bmatrix} 0.1432 & 0.3580 \\ 0.5012 & 0.7160 \end{bmatrix} + \begin{bmatrix} 0.1398 & -0.1398 \\ 0.0699 & -0.0699 \end{bmatrix} \\ x_3 &= \begin{bmatrix} 0.2830 & 0.2182 \\ 0.5711 & 0.6461 \end{bmatrix}\end{aligned}$$

- b) Explain how Stable Diffusion generates an image from a text prompt. In your answer, describe the role of the text encoder, latent space, UNet, scheduler, text conditioning, and autoencoder decoder. Also draw a neat diagram of the Stable Diffusion pipeline.

Answer

Stable Diffusion generates images by performing the diffusion process in a compressed latent space rather than directly in pixel space. This makes the model more efficient because denoising high-resolution images in pixel space requires large memory and computation.

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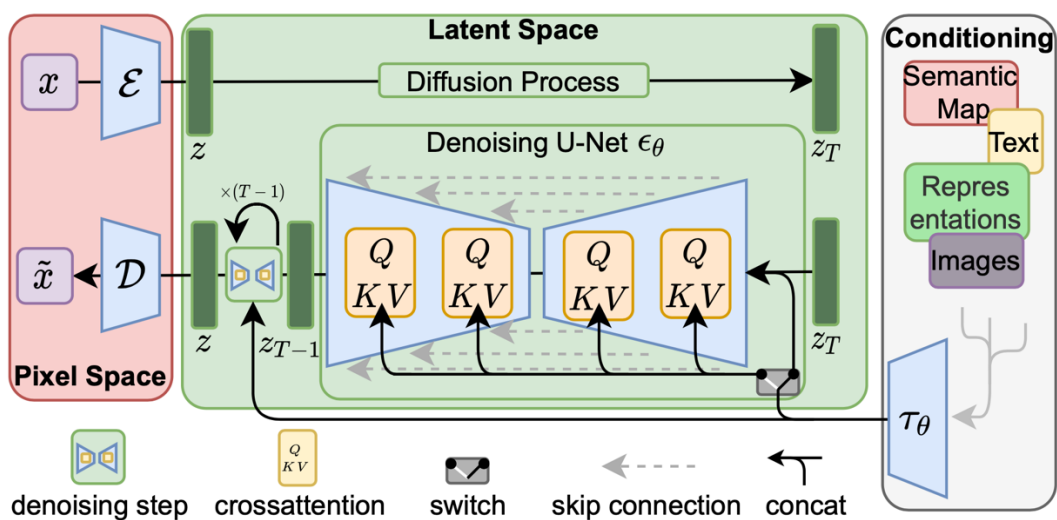
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The process begins with a text prompt. The prompt is passed through a text encoder, commonly CLIP-based, which converts the input text into text embeddings. These embeddings represent the semantic meaning of the prompt and guide the image generation process.

Stable Diffusion starts from random noise in latent space. The UNet receives this noisy latent representation, the diffusion timestep, and the text embeddings. The scheduler controls how the noisy latent is updated at each denoising step. At each step, the UNet predicts the noise that should be removed from the latent representation.

Text conditioning is used to guide the denoising process toward the meaning of the prompt. This is commonly achieved through attention mechanisms, where the latent image representation uses information from the text embeddings. As a result, the generated image becomes aligned with the prompt.

After several denoising steps, the model obtains a cleaner latent representation. This final latent representation is passed through the autoencoder decoder, which converts it from latent space into pixel space. The output is the final generated image.



Question No 5. [5+5]

- a) With the help of a diagram explain the Retrieval-Augmented Generation (RAG) pipeline. Describe each component of the pipeline. Additionally, discuss how RAG helps mitigate limitations of standalone LLMs. Write one disadvantages of RAG based systems.

Retrieval-Augmented Generation (RAG) Pipeline

Retrieval-Augmented Generation (RAG) is a framework that combines information retrieval with large language models to generate responses grounded in external knowledge sources. As illustrated in the diagram shown in class, the RAG pipeline is composed of three main stages: ingestion, retrieval, and synthesis.

Ingestion Phase

In the ingestion phase, raw documents such as PDFs, reports, web pages, or domain-specific files are collected and prepared for retrieval. They are divided into smaller chunks to make semantic matching more effective and to fit within model context limits. Each chunk is then converted into a dense vector representation using an embedding model that captures its semantic meaning. These embeddings are stored in a vector index.

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Retrieval Phase

In the retrieval phase, the query is first transformed into an embedding using the same embedding model. This query embedding is then compared against the stored document embeddings in the index using a similarity metric such as cosine similarity. Based on this comparison, the system retrieves the top-K most relevant document chunks that are semantically closest to the query.

Synthesis Phase

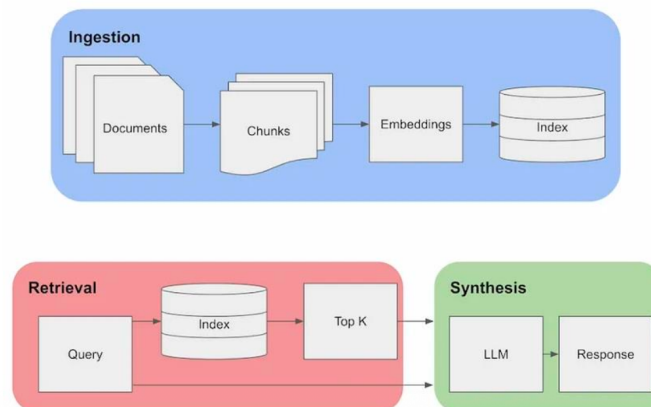
In the synthesis phase, the retrieved document chunks are injected into the prompt of the large language model along with the user query. The language model then generates a response by synthesizing its parametric knowledge with the retrieved contextual information. This ensures that the generated output is more accurate, context-aware, and aligned with the external data, reducing the likelihood of unsupported or fabricated responses.

Mitigating Limitations of Standalone LLMs

- RAG helps address several limitations of standalone large language models by providing access to external knowledge at inference time.
- Because responses are grounded in retrieved documents, the risk of hallucination is significantly reduced and factual accuracy is improved.
- RAG also allows the system to incorporate up-to-date and domain-specific information without retraining the model, making it more flexible and cost-effective.

Disadvantages of RAG-Based Systems

- RAG-based systems introduce additional complexity compared to standalone language models.
- The system requires multiple components such as document preprocessing pipelines, embedding models, vector indexes, and retrieval mechanisms, which increases engineering and maintenance overhead.
- Retrieval adds latency to inference, and the overall quality of responses heavily depends on the effectiveness of chunking and retrieval strategies.
- The limited context window of language models restricts how much retrieved information can be used at once, and maintaining an up-to-date and consistent index requires continuous effort.



- b) Explain adversarial distillation, and cross-modal distillation. Discuss the main idea of each method, and one suitable use case for each.

Answer

Adversarial distillation uses an adversarial training setup to improve the student model. A discriminator or adversarial objective is used to make the student's outputs or representations closer to the teacher's. The goal is not only to match predictions, but also to make the student produce

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teacher-like behavior that is difficult to distinguish from the teacher. It is useful when the student must closely imitate the teacher's output distribution or feature representation.

Cross-modal distillation transfers knowledge between models working on different data modalities. For example, a vision-language model may guide a text-only model, or an image-based teacher may transfer visual knowledge to a smaller student model. The main idea is to use knowledge from one modality to improve learning in another modality. It is useful when one modality has richer supervision or a stronger model, but the deployed student model must operate with a different or limited modality.